

Experiment 2 - Write a Python program to solve the 8-Queens Problem.

Program Code

```
def is_safe(board, row, col):
    for i in range(row):
        if board[i] == col:
            return False

        if abs(board[i] - col) == abs(i - row):
            return False

    return True

def solve_queens(board, row):
    if row == 8:
        return True

    for col in range(8):
        if is_safe(board, row, col):
            board[row] = col

            if solve_queens(board, row + 1):
                return True

            board[row] = -1

    return False
```

```
def print_board(board):
    for row in range(8):
        for col in range(8):
            if board[row] == col:
                print("Q", end=" ")
            else:
                print(".", end=" ")
        print()
```

```
board = [-1] * 8
```

```
if solve_queens(board, 0):
    print("Solution found!")
    print()
    print_board(board)
else:
    print("No solution found.")
```

Program Output

```
PS D:\AI LAB> & C:\Users\Student\AppData\Local\Programs\Python\Python313\python.exe "d:/AI LAB/experiment2_8_queens.py"
Solution found!

Q . . . . .
. . . . Q . .
. . . . . Q
. . . . Q . .
. . Q . . . .
. . . . . Q .
. Q . . . . .
. . . Q . . .
PS D:\AI LAB> 
```